

The effect of negative feedback loops on the dynamics of Boolean networks¹

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Abstract

Feedback loops play an important role in determining the dynamics of biological networks. In order to study the role of negative feedback loops, this paper introduces the notion of “distance to positive feedback (PF-distance)” which in essence captures the number of “independent” negative feedback loops in the network, a property inherent in the network topology. Through a computational study using Boolean networks, it is shown that PF-distance has a strong influence on network dynamics and correlates very well with the number and length of limit cycles in the phase space of the network. To be precise, it is shown that, as the number of independent negative feedback loops increases, the number (length) of limit cycles tends to decrease (increase). These conclusions are consistent with the fact that certain natural biological networks exhibit generally regular behavior and have fewer negative feedback loops than randomized networks with the same number of nodes and same connectivity.

Key words: Feedback loops; Boolean networks; gene regulatory networks; limit cycles.

Introduction

An understanding of the design principles of biochemical networks, such as gene regulatory, metabolic, or intracellular signaling networks is a central concern of systems biology. In particular, the intricate interplay between network topology and resulting dynamics is crucial to our understanding of such networks, as is their presumed modular structure. Features that relate network topology to dynamics may be considered “robust” in the sense that their influence does not depend on detailed quantitative features such as exact flux rates. A topological feature of central interest in this context is the existence of positive and negative feedback loops. There is broad consensus that feedback loops have a decisive effect on dynamics, which has been studied extensively through the analysis of mathematical network models, both continuous and discrete. Indeed, it has long been appreciated by biologists that positive and negative feedback loops play a central role in controlling the dynamics of a wide range of biological systems. Thomas et al. (1) conjectured that positive feedback loops are necessary for multistationarity whereas negative feedback loops are necessary for the existence of periodic behaviors. Proofs for different partial cases of these conjectures have been given, see (2–6). Moreover, it has often been pointed out (see, for instance, (7)) that an abundance of loops, and specifically negative loops (8, 9), should result in longer cycles and thus more “chaotic” behavior in the network. Our results provide strong evidence in support of this.

We focus here on Boolean network (BN) models, a popular model type for biochemical networks, initially introduced by Kauffman (10). In particular, we study BN models in which each directed edge can be characterized as either an inhibition or an activation. In Boolean models of biological networks, each variable can only attain two values (0/1 or “on/off”). These values represent whether a gene is being expressed, or the concentration of a protein is above a certain threshold, at time t . When detailed information on kinetic rates of protein-DNA or protein-protein interactions is lacking, and especially if regulatory relationships are strongly sigmoidal, such models are useful in theoretical analysis, because they serve to focus attention on the basic dynamical characteristics while ignoring specifics of reaction mechanisms, see (11–14).

Boolean networks constructed from monotone Boolean functions (i.e. each node or “gate” computes a function which is increasing on all arguments) are of particular interest, and have been studied extensively, in the electronic circuit design and pattern recognition literature (15, 16), as well

as in the computer science literature; see e.g. (17–19) for recent references. For Boolean and all other finite iterated systems, all trajectories must either settle into equilibria or into periodic orbits, whether the system is made up of monotone functions or not, but monotone networks have always somewhat shorter cycles. This is because periodic orbits must be anti-chains, i.e. no two different states can be compared; see (15, 20). An upper bound may be obtained by appealing to Sperner’s Theorem (21): Boolean systems on n variables can have orbits of period up to 2^n , but monotone systems cannot have orbits of size larger than $\binom{n}{\lfloor n/2 \rfloor} \approx 2^n \sqrt{\frac{2}{n\pi}}$; these are all classical facts in Boolean circuit design (15). It is also known that the upper bound is tight (15), in the sense that it is possible to construct Boolean systems on n variables, made up of monotone functions, for which orbits of the maximal size $\binom{n}{\lfloor n/2 \rfloor}$ given by Sperner’s Theorem exist. This number is still exponential in n . However, anecdotal experience suggests that monotone systems constructed according to reasonable interconnection topologies and/or using restricted classes of gate functions, tend to exhibit shorter orbits (22, 23). One may ask if the *architecture* of the network, that is, the structure of its dependency (also called interconnection) graph, helps insure shorter orbits. In this direction, Aracena et al. (17) showed that on certain graphs, called there “caterpillars,” monotone networks can only have cycles of length at most two in their phase space.

The present paper asks the even more general question of whether networks that are not necessarily made up from monotone functions, but which are “close to monotone” (in a sense to be made precise, roughly meaning that there are few independent negative loops) have shorter cycles than networks which are relatively farther to monotone.

In (8), we conjectured that “smaller distance to monotone” should correlate with more ordered (less “chaotic”) behavior, for random Boolean networks. A partial confirmation of this conjecture was provided in (9), where the relationship between the dynamics of random Boolean networks and the ratio of negative to positive feedback loops was investigated, albeit only for the special case of small Kauffman-type NK and NE networks, and with the additional restriction that all nodes have the same function chosen from AND, OR, or UNBIAS. Based on computer simulations, the authors of (9) found a positive (negative) correlation between the ratio of fixed points (other limit cycles) and the ratio of positive feedback loops. Observe that this differs from our conjecture in two fundamental ways: (i) our measure of disorder is related to the number of “independent” negative loops, rather than their absolute number, and (ii) we do not consider that the number

of positive loops should be part of this measure: a large number of negative loops will tend to produce large periodic orbits, even if the negative to positive ratio is small due to a larger number of positive loops.

Thus, in the spirit of the conjecture in (8), the current paper has as its goal an experimental study (as opposed to a theoretical analysis) of the effect of independent negative feedback loops on network dynamics, based on an appropriately defined measure of *distance to positive-feedback*. We study the effect of this distance on features of the network dynamics, namely the number and length of limit cycles. Rather than focusing on the number of negative feedback loops in the network as the characteristic feature of a network, we focus on the number of switches of the activation/inhibition character of edges that need to be made in order to obtain a network that has only positive feedback loops. We relate this measure to the cycle structure of the phase space of the network. It is worth emphasizing that the absolute number of negative feedback loops and the distance to positive feedback are not correlated in any direct way, as it is easy to construct networks with a fixed distance to positive feedback that have arbitrarily many negative feedback loops, see Figure 1.

Motivations

There are three different motivations for posing the question that we ask in this paper. The first is that most biological networks appear to have highly regular dynamical behavior, settling upon simple periodic orbits or steady states. The second motivation is that it appears that real biological networks such as gene regulatory networks and protein signaling networks are indeed close to monotone (24–26). Thus, one may ask if being close to monotone correlates in some way with shorter cycles. Unfortunately, as mentioned above, one can build networks that are monotone yet exhibit exponentially long orbits. This suggests that one way to formulate the problem is through a statistical exploration of graph topologies, and that is what we do here.

A third motivation arises from the study of systems with continuous variables, which arguably provide more accurate models of biochemical networks. There is a rich theory of continuous-variable monotone (to be more precise, “cooperative”) systems. These are systems defined by the property that an inequality $\mathbf{a}(0) < \mathbf{b}(0)$ in initial conditions propagates in time so that the inequality $\mathbf{a}(t) < \mathbf{b}(t)$ remains true for all future times $t > 0$. Note that this is entirely analogous to the Boolean case, when one makes the obvious definition that two Boolean vectors satisfy the inequality $\mathbf{a} = (a_1, \dots, a_n) \leq \mathbf{b} = (b_1, \dots, b_n)$ if $a_i \leq b_i$ for each $i = 1, \dots, n$ (setting

$0 < 1$). Monotone continuous systems have convergent behavior. For example, in continuous-time (ordinary differential models), they cannot admit any possible stable oscillations (27–29), and, when there is only one steady state, every bounded solution converges to this unique steady state (monostability), see (30). When, instead, there are multiple steady-states, the Hirsch Generic Convergence Theorem is the fundamental result (20, 29, 31, 32); it states, under an additional technical assumption (“strong” monotonicity) that generic bounded solutions must converge to the set of steady states. Biological applications of these theorems include positive gene feedback circuits (20) as well as single phosphorylation/dephosphorylation loops (33) and double phosphorylation/dephosphorylation loops under appropriate assumptions on kinetic constants (34). For discrete-time strongly monotone systems, generically also stable oscillations are allowed besides convergence to equilibria, but no more complicated behavior. In neither case, discrete-time or continuous-time continuous monotone systems, does one observe “chaotic” behavior. It is an open question whether continuous systems that are in some sense close to being monotone have more regular behavior, in a statistical sense, than systems that are far from being monotone, just as for the Boolean analog considered in this paper. The Boolean case is more amenable to computational exploration than continuous-variable systems, however. Since long orbits in discrete systems may be viewed as an analog of chaotic behavior, we focus on lengths of orbits.

One can proceed in several ways to define precisely the meaning of distance to positive feedback. One associates to a network made of unate (definition below) gate functions a signed graph whose edges have signs (positive or negative) that indicate how each variable affects each other variable (activation or inhibition). The first definition, explored in (8, 24, 25, 35, 36) starts from the observation that in a network with all monotone node functions there are no negative undirected cycles. Conversely, if the dependency graph has no undirected negative parity cycles (a “sign-consistent” graph), then a change of coordinates (globally replacing a subset of the variables by their complements) renders the overall system monotone. Thus, asking what is the smallest number of sign-flips needed to render a graph sign-consistent is one way to define distance to monotone. This approach makes contact with areas of statistical physics (the number in question amounts to the ground energy of an associated Ising spin-glass model), as well as with the general theory of graph-balancing for signed graphs (37) that originated in (38). It is also consistent with the generally accepted meaning of “monotone with respect to some orthant order” in the ODE literature as a system that is cooperative under some inversion of variables.

A second, and different, definition starts from the fact that a network with all monotone node functions has, in particular, no negative-sign *directed* loops. For a strongly connected graph, the property that no directed negative cycles exist is equivalent to the property that no undirected negative cycles exist. However, for non-strongly connected graphs, the properties are not the same. Thus, this second property is weaker. The second property is closer to what biologists and engineers mean by “not having negative feedbacks” in a system, and hence is perhaps more natural for applications. In addition, it is intuitively clear that negative feedbacks should be correlated to possible oscillatory behavior. (This is basically Thomas’ conjecture. See (25) for precise statements for continuous-time systems; interestingly, published proofs of Thomas’ conjecture use the first definition, because they appeal to results from monotone dynamical systems.) Thus, one could also define distance to monotone as the smallest number of sign-flips needed to render a graph free of negative directed loops. To avoid confusion, we will call this notion, which is the one studied in this paper, *distance to positive-feedback*, or just “PF-distance.”

Theory

Distance to positive-feedback

We give here the basic definitions of the concepts relevant to the study.

Definition 1 *Let $k = \{0, 1\}$ be the field with two elements. We order the two elements as $0 < 1$. This ordering can be extended to a partial ordering on k^n by comparing vectors coordinate-wise in the lexicographic ordering.*

1. *A Boolean function $h : k^n \rightarrow k$ is monotone if, whenever $\mathbf{a} \leq \mathbf{b}$ coordinate-wise, for $\mathbf{a}, \mathbf{b} \in k^n$, then $h(\mathbf{a}) \leq h(\mathbf{b})$.*
2. *A Boolean function h is unate if, whenever x_i appears in h , the following holds: Either*
 - (a) *For all $a_1, \dots, a_{i-1}, a_{i+1}, \dots, a_n \in k$,*

$$h(a_1, \dots, a_{i-1}, 0, a_{i+1}, \dots, a_n) \leq h(a_1, \dots, a_{i-1}, 1, a_{i+1}, \dots, a_n),$$
or
 - (b) *For all $a_1, \dots, a_{i-1}, a_{i+1}, \dots, a_n \in k$,*

$$h(a_1, \dots, a_{i-1}, 0, a_{i+1}, \dots, a_n) \geq h(a_1, \dots, a_{i-1}, 1, a_{i+1}, \dots, a_n).$$

The definition of unate function is equivalent to requiring that whenever a_i appears in h , then it appears either everywhere as a_i or everywhere as $\neg a_i := 1 + a_i$.

Let f be a Boolean network with variables $x_1, \dots, x_n$, and coordinate functions $f_1, \dots, f_n$. That is, $f = (f_1, \dots, f_n) : k^n \rightarrow k^n$. We can associate to f its *dependency graph* $\mathcal{D}(f)$: The vertices are $v_1, \dots, v_n$, corresponding to the variables $x_1, \dots, x_n$, and there is an edge $v_i \rightarrow v_j$ if and only if x_i appears in f_j . If all coordinate functions f_i of f are unate, then the dependency graph of f is a signed graph. Namely, we associate to an edge $v_i \rightarrow v_j$ a “+” if f_j preserves the ordering as in 2(a) of Definition 1 and a “-” if it reverses the ordering as in 2(b) of Definition 1. For later use we observe that this graph (as any directed graph) can be decomposed into a collection of strongly connected components, with edges between strongly connected components going one way but not the other. (Recall that a strongly connected directed graph is one in which any two vertices are connected by a directed path.) That is, the graph can be represented by a partially ordered set in which the strongly connected components make up the elements and the edge direction between components determines the order in the partially ordered set.

Definition 2 *Let f be a Boolean network with unate Boolean functions and $\mathcal{D}(f)$ be its signed dependency graph. Then*

1. *The graph $\mathcal{D}(f)$ is a positive-feedback (PF) if it does not contain any odd parity directed cycles. (The parity of a directed cycle is the product of the signs of all the edges in the cycle.) In this case, f is called a positive-feedback network (PF).*
2. *The PF-distance of f is the smallest number of signs that need to be changed in the dependency graph to obtain a PF network. We denote this number by $|\mathcal{D}(f)|$ or simply $|f|$.*

Notice that for a given directed graph G , different assignments of sign to the edges produce graphs with varying PF-distance. In particular, there is a maximal PF-distance that a given graph topology can support.

The dynamics of f are presented in a directed graph, called the *phase space* of f , which has the 2^n elements of k^n as a vertex set, and there is an edge $\mathbf{a} \rightarrow \mathbf{b}$ if $f(\mathbf{a}) = \mathbf{b}$. It is straightforward to see that each component of the phase space has the structure of a directed cycle, a *limit cycle*, with a directed tree feeding into each node of the limit cycle. The elements of these trees are called *transient states*.

In this paper we relate the dynamics of a Boolean network to its PF-distance. The following is a motivational example that explains the main results.

Example 3 Let G be the directed graph depicted in Figure 2 (left). It is easy to check that the maximal PF-distance of G is 3. Let $f = (x_3 \vee \neg x_4, x_1 \wedge x_2, x_2 \wedge \neg x_4, \neg x_3) : \{0, 1\}^4 \longrightarrow \{0, 1\}^4$ and $g = (\neg x_3 \vee x_4, x_1 \wedge \neg x_2, x_2 \wedge x_4, \neg x_3) : \{0, 1\}^4 \longrightarrow \{0, 1\}^4$. It is clear that f and g are sign-modifications of the same PF network $(x_3 \vee x_4, x_1 \wedge x_2, x_2 \wedge x_4, x_3)$, in particular, they have the same (unsigned) dependency graph. However, the PF-distance of f is 0 while it is 3 for g . The phase space of f is depicted in Figure 2 (middle) and that of g is on the right. Notice that f has two limit cycles of lengths 1 and 2, respectively, while g has only one limit cycle of length 4.

For each distance $0 \leq d \leq 3$, we analyze the dynamics of 10 random PF networks and their sign modifications of distance d on the directed graph in Figure 2 (left). The average of the numbers (lengths) of limit cycles is computed as in Table 1. The best fit-line of the averages of the number (length) of limit cycles is computed and its slope is reported as in Figure 3. The details of this analysis are provided in the Supporting Information.

Note that the purpose of the experiment is to determine a trend in the number and length of cycles as the PF-distance increases. It is thus appropriate to use a straight line to approximate the data points, even though this might not at all be the best possible approximation. But the slope of the line of best fit incorporates that trend adequately. We have repeated the experiment in Example 3 above for many different graphs and observed that the slope of the best fit-line of the length (resp. number) of limit cycles is positive (resp. negative) most of the time. In the next section we present the details of the experiments and the algorithms used in the computations.

Methods

The main results of this paper relate the PF-distance of Boolean networks with the number and length of their limit cycles. Specifically, our hypothesis is that, for Boolean networks consisting of unate functions, *as the PF-distance increases, the total number of limit cycles decreases on average and their average length increases*. This is equivalent to saying that for most or all experiments this slope is negative for the number of limit cycles and is positive for their length.

To test this hypothesis we analyzed the dynamics of more than six million Boolean networks arranged in about 130,000 experiments.

Random generation of unate functions

We generated a total of more than 130,000 random directed graphs, where each graph has 5, 7, 10, 15, 20, or 100 nodes, with maximum in-degree 5 for each node. The graphs were generated as random adjacency matrices, with the restriction that each row has at least one 1 and at most five 1's. For each directed graph G , we generated 10 Boolean networks with unate functions and dependency graph G , by using the following fact.

Lemma 4 *A Boolean function f of n variables is unate if and only if it is of the form $f(\mathbf{x}) = g(\mathbf{x} + \mathbf{s})$, where g is a monotone Boolean function of n variables and $\mathbf{s} \in k^n$ and “+” denotes addition modulo 2.*

Proof. If f is unate then each variable x_i appears in f always as x_i or always as $\neg x_i$. Suppose that all x_i appear without negations. Then f is constructed using $\wedge$ and $\vee$. Hence f is monotone. Otherwise, let $\mathbf{s} \in k^n$ be the vector whose i th entry is 1 if and only if x_i appears as $\neg x_i$ in f . Then $g(\mathbf{x}) = f(\mathbf{x} + \mathbf{s})$ is a monotone function and $f(\mathbf{x}) = g(\mathbf{x} + \mathbf{s})$. The converse is clear.

So in order to generate unate functions it is sufficient to generate monotone functions. We generated the set M_i of monotone functions in i variables by exhaustive search for $i = 1, \dots, 5$. (For example, M_5 has 6894 elements.) Unate functions for a given signed dependency graph can then be generated by choosing random functions from M_i and random vectors $\mathbf{s} \in k^n$. The nonzero entries in $\mathbf{s}$ for a given node correspond to the incoming edges with negative sign in the dependency graph. Using this process we generated Boolean networks with unate Boolean functions.

We then carry out the following steps.

The Experiment

Let G be a random unsigned directed graph on n nodes with a maximal PF-distance t , and let $D \leq t$. Consider 10 unate Boolean networks chosen at random with G as their dependency graph.

1. For $1 \leq d \leq D$, let G_d be a signed graph of G of distance d .
 - (a) For each network f of the ten networks,
 - i. Let g be a modified network of f such that $\mathcal{D}(g) = G_d$; the signed dependency graph of g is G_d .

- ii. Compute the number and length of all limit cycles in the phase space of g .
- (b) Compute the average number N (resp. average length L) of limit cycles in the phase spaces of the $g's$.
- 2. Compute the slope s_N (resp. s_L) of the best fit-line of the $N's$ (resp. $L's$).

The output of a single experiment consists of the two non-negative integers s_N and s_L .

Computation of PF-distance

Let f be a Boolean network with unate Boolean functions and let $|f|$ be its PF-distance. The proofs of the following facts are straightforward.

1. Suppose the dependency graph of f has a negative feedback loop at a vertex. Let f' be the Boolean network obtained by changing a single sign to make the loop positive. Then $|f| = |f'| + 1$.
2. Let $H_1, \dots, H_s$ be the strongly connected components of the dependency graph $D(f)$. Then

$$|\mathcal{D}(f)| = \sum_{i=1}^s |H_i|.$$

The algorithm for computing $|H_i|$ now follows.

Algorithm: Distance to PF

Input: A signed, directed, and strongly connected graph G .

Output: $|G|$; the PF-distance of G .

Let $d = 0$.

Step 1: Let $G_1, \dots, G_r$ be the collection of all signed graphs obtained by making exactly d sign changes in G .

Step 2: For $i = 1, \dots, r$
If G_i is PF, then RETURN $|G| = d$.

Step 3: Otherwise, $d := d + 1$, Go to Step 1 above.

In Step 2 above, to check whether a strongly connected graph is PF, it is equivalent to check whether it has any (undirected) negative cycles, which can easily be done in many different ways, see, e.g., (25). This algorithm must terminate, since G has finitely many edges and hence the PF-distance of G is finite.

If G has m directed edges, then there are 2^m possible sign assignments. However, to compute the maximal PF-distance, one does not need to find the PF-distance of such possible assignments, see *Supporting Information* for the algorithm we used to compute the maximal PF-distance.

We have coded these algorithms into a *Mathematica* code that we use to carry out our Experiment. Once the number of nodes n , the maximal in-degree k of each node, and the maximal considered distance D are fixed, a random directed graph is generated and the experiment is carried out as shown above. Although we will not analyze the complexity here, it is clear that, as the value of n , k or D increases, the time needed to do the Experiment increases exponentially. In Table 2 we report the average cpu time (minutes) for some of the experiments where two of the variables are fixed.

Results

As mentioned in the previous section, we have carried out approximately 130,000 such experiments for networks whose number of nodes ranges from 5 to 100. Table 3 summarizes the outcome of these experiments and represents the main result of the paper. The right-most two columns list the percentage of experiments for each network size that conform to our main hypothesis. As can be seen, the table confirms the hypothesis strongly, in particular for networks of smaller size. In order to explain the drop in the percentage of networks that conform to the hypothesis for larger networks, we need to consider some details of the experiments.

To begin with, we need to observe that a computationally expensive part of an experiment is the computation of the maximal PF-distance which a given directed graph topology can support, since this is done essentially by an exhaustive test of all possible sign distributions on the edges. This computation becomes prohibitive for even modest-size graphs, with, e.g., 10 nodes. So unlike in Example 3, for networks on more than 5 nodes, we only considered PF-distances that are less than or equal to the number of nodes in the network. (See the Methods Section for a detailed description of the experiment.) In fact, for graphs with 20 (resp. 100) nodes, all considered

networks have PF-distance less than or equal to 5 (resp. 10). That is, our experiments reflect only the relationship between dynamics and PF-distance for networks that are very close to positive feedback, relative to their actual PF-distance.

For networks on 5 nodes, we carried out 4000 experiments by varying the PF-distance considered in the computations. Table 4 shows the number of experiments that do not conform to our hypothesis as we vary the considered PF-distance. A more instructive way to visualize the effect of restricting the range of PF-distance relative to maximal PF-distance on the likelihood of experiments validating the conjecture is in the form of a histogram. In Figures 4 and 5, the horizontal axis represents the slope of the lines of best fit and the vertical axis represents the percentage of experiments that confirm our hypothesis. Figure 4 shows the results of the 4000 experiments on 5-node networks. The histograms show the results when PF-distance of the network is 25%, resp. 50%, resp. 75%, resp. 100% of the maximal PF-distance. It can be seen as the allowed range of PF-distance approaches the maximal distance, almost all experiments show positive slope of the best-fit line, thereby conforming to the conjecture. Similar results for the average number of limit cycles are shown in Figure 5, demonstrating that if the PF-distance of networks is allowed the whole possible range, almost all the experiments conform to our hypothesis as we already noticed in Table 4.

We have carried out similar computations for networks with 7 (5000 experiments) and 10 (6000 experiments) nodes; see the supporting information for details. The results there are not quite as clear as for 5 node networks, primarily because all computations were done with PF-distance less than or equal to 5, due to the computational complexity involved. For instance, for networks with 10 nodes (and up to 4 incoming edges per node) 31 out of 1000 experiments did not conform to our hypothesis for networks with PF-distance up to 5.

In summary, the extensive computations confirm our hypothesis that, as the PF-distance increases, the total number of limit cycles decreases on average and their average length increases. Furthermore, the slopes of the best-fit lines increasingly conform to our hypothesis the closer the PF-distance of the networks comes to the maximum PF-distance of the network topology.

Supporting Information

For details of our analysis of the 5-node networks and all other considered networks, see *Supporting Text*.

Discussion

Negative feedback loops in biological networks play a crucial role in controlling network dynamics. The new measure of “distance to positive feedback (PF-distance)” introduced in this paper is designed to capture the notion of “independent” feedback loops. We have shown that PF-distance correlates very well with the average number and length of limit cycles in networks, key measures of network dynamics. By analyzing the dynamics of more than six millions Boolean networks, we have provided evidence that networks with a larger number of independent negative feedback loops tend to have longer limit cycles and thus may exhibit more “random” or “chaotic” behavior. Furthermore, the number of limit cycles tends to decrease as the number of independent negative feedback loops increases.

In general, the problem of computing the PF-distance of a network is NP-complete, as MAX-CUT can be mapped into it as a special case; see (24, 25, 35) for a discussion for the analogous problem of distance to monotone. The question of computing *distance to monotone* has been the subject of a few recent papers (24, 35, 36). The first two of these proposed a randomized algorithm based on a semi-definite programming relaxation, while the last one suggested an efficient deterministic algorithm for graphs with small distance to monotone. Since a strongly connected component of a graph is monotone if and only if it has the PF property, methods for computing PF distance for large graphs may be developed by similar techniques. Work along these lines is in progress.

One may speculate that the regular behavior observed in biological networks is due, in some measure, to their presumably small distance to positive feedback. There are as of yet too few large networks known with information on directionality and signs of interactions, to be able to draw statistically meaningful conclusions. However, one may make some preliminary statements. It was shown in (7) that for five networks (*E. coli* transcription, *S. cerevisiae* transcription, STKE signaling, *Drosophila* transcription, and a *C. elegans* neuron), feedback loops are statistically highly underrepresented among other motifs. No sign information was available for some of these networks, but it has been observed that negative (“incoherent”) cycles are less abundant than positive (“coherent”) ones in certain networks (39), and this will in turn bias feedback loops to be negative, when post-transcriptional modifications are added to the model. Similarly, models of a segment polarity network in *Drosophila* and of an *S. cerevisiae* gene network were shown to be close to monotone (no negative undirected cycles) compared to random graphs (24, 25, 35), and a similar statement of small PF distance

(now including feedback loops as well as feedforward loops) was made for a CA-1 neuron signaling network and an *E. coli* and *S. cerevisiae* network (26), although this study was restricted to small loops.

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Figure Legends

Figure 1.

A graph that has an arbitrary number of negative loops, as many as the number of nodes in the second layer, but PF distance one: to avoid negative feedback, it suffices to switch the sign of the single (negative) arrow from the bottom to the top node. All unlabeled arrows are positive.

Figure 2.

The dependency graph (left), the phase space of f (middle) and the phase space of g (right) from Example 3. These graph were generated using DVD (40).

Figure 3.

The best fit-line of the averages of the numbers (lengths) of limit cycles from Table 1.

Table 1.

The average of the numbers (lengths) of limit cycles of the networks from Example 3.

Table 2.

The average cpu time (seconds) as we increase the number of nodes n , the maximum in-degree of each node k , and the maximal-considered distance D .

Table 3.

The percentage of experiments that conform the hypotheses.

Table 4.

The number of experiments that did not conform to our hypothesis for 5-node networks. We considered PF-distance 25%, 50%, 75%, and 100% of the maximal distance. For each d , we considered 1000 experiments.

Figure 4.

5-node networks. Histogram of slopes of best-fit lines to the average length of limit cycles (horizontal axis) vs. percentage of experiments with a given slope (vertical axis). The panels from left to right include networks with increasing PF-distance, with 25%, 50%, 75%, and 100% of the maximal distance.

Figure 5.

5-node networks. Histogram of slopes of best-fit lines to the average number of limit cycles (horizontal axis) vs. percentage of experiments with a given slope (vertical axis). The panels from left to right include networks with increasing PF-distance, with 25%, 50%, 75%, and 100% of the maximal distance.

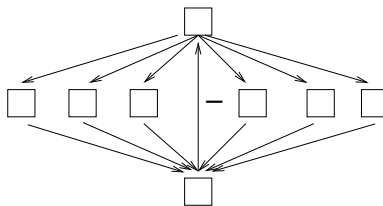


Figure 1:

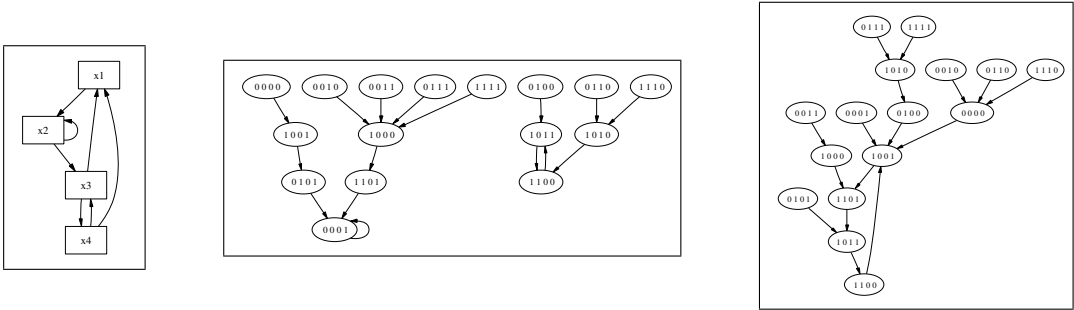


Figure 2:

Table 1:

d	Av. Num.	Av. Len.
0	3.5	1.23
1	2.80	1.25
2	2.50	1.52
3	1.20	3.50

Table 2:

$k = 2$ and $D = 4$		$n = 5$ and $D = Max.$		$n = 10$ and $k = 5$	
n	Av. cpu	k	Av. cpu	D	Av. cpu
5	.5951	2	.6256	5	56.779
7	2.842	3	1.525	6	82.718
10	32.892	4	7.4078	7	101.35
15	2445.3	5	208.39	8	204.45

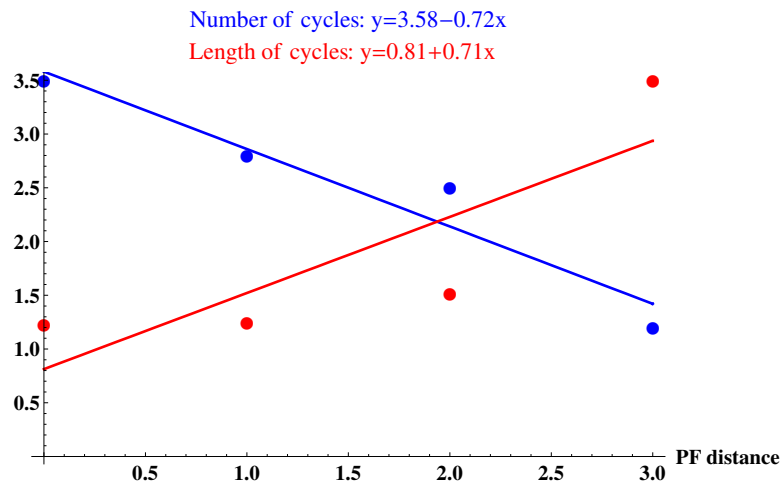


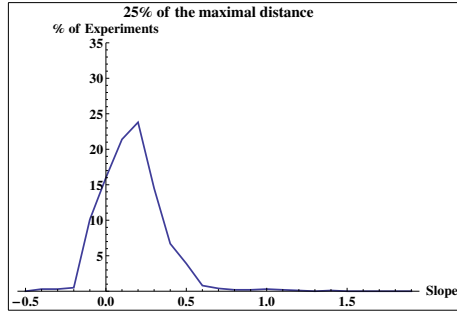
Figure 3:

Table 3:

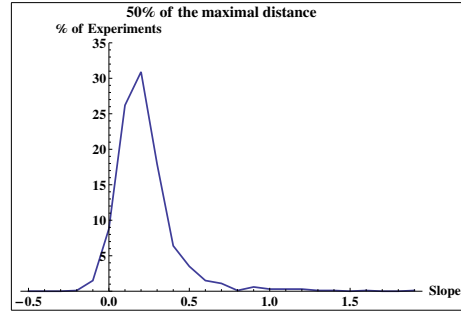
n	Num. of Exp.	Av. Num.	Av. Len.
5	117000	99.75	99.83
7	5000	97.82	99.92
10	6000	95.70	99.58
15	2921	95.72	98.25
20	331	90.03	94.86
100	659	77.39	93.93

Table 4:

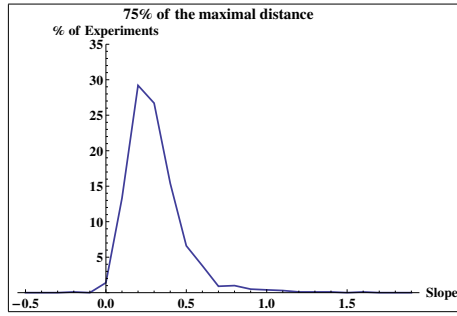
D	Av. Num.	Av. Len.	Med. Num.	Med. Len.
25%	26	114	29	542
50%	4	16	18	59
75%	0	1	6	3
100%	1	0	2	0



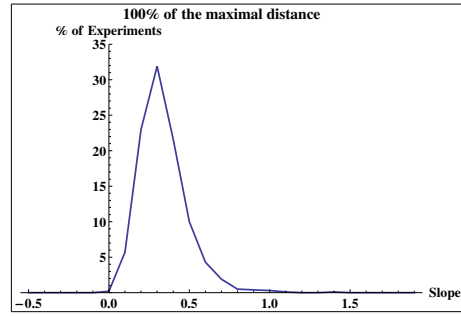
(a) 25% of the maximal distance



(b) 50% of the maximal distance



(c) 75% of the maximal distance



(d) 100% of the maximal distance

Figure 4:

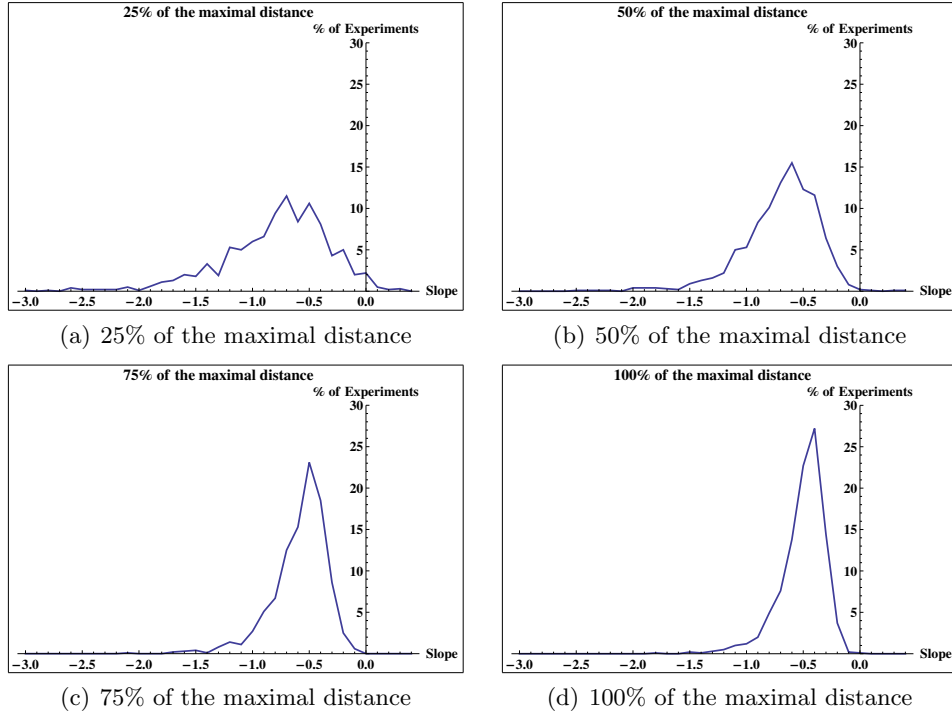


Figure 5:

Supporting Information

1. Motivational Example.

Consider the directed graph G in Figure 1, it is not hard to check that its maximal PF-distance is 3. At random, choose ten Boolean networks such that all have the same dependency graph G . For example, consider the Boolean network $f(x_1, x_2, x_3, x_4) = (x_3 \vee x_4, x_1 \wedge x_2, x_2 \wedge x_4, x_3)$.

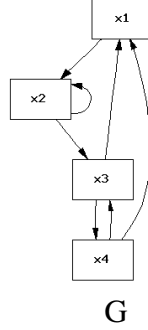


Figure 1. A directed graph on 4 nodes.

Let G_0, G_1, G_2, G_3 be signed graphs of G of distance 0,1,2,3, respectively, as in Figure 2. Let $S = \{G_0, G_1, G_2, G_3\}$. For each G_d , by replacing x_i by $\sim x_i$ wherever it appears for some i , modify f into f' such that G_d is the dependency graph of f' , see Figure 2. Similarly, we modify each of the other 9 networks.

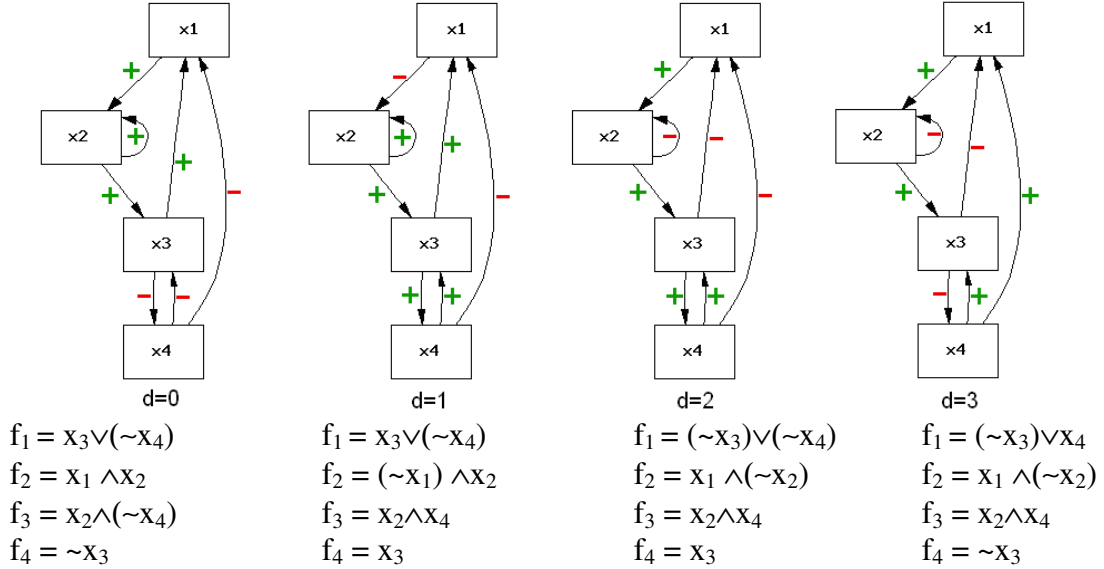


Figure 2. Possible sign assignments of the graph G and the corresponding modified network of f for each signed graph. For example, for $d=1$, since the are negative edges from x_4 to x_1 and x_1 to x_2 , we write $\sim x_4$ in f_1 and $\sim x_1$ in f_2

For each d , we find the phase spaces of the 10 Boolean networks that have G_d as their signed dependency graph.

Compute the average number as well as average length of limit cycles. We summarize this in Table 1.

d	0	1	2	3
Average number of cycles	3.50	2.80	2.50	1.20
Average Length of cycles	1.23	1.25	1.52	3.50

Table 1. The average number and average length of limit cycles with respect to d . For example, for networks with PF distance 0, the average number of cycles is 3.50 and their average length is 1.23.

We plot the average as a function of d and compute the slope of the best fit line. In our example, the slope of the best fit line of the numbers is -0.72 and the slope of the best fit line of the lengths is 0.71.

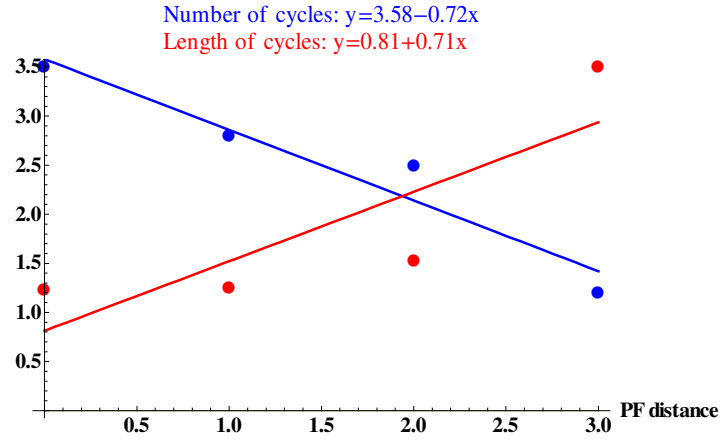


Figure 3. The best fit-line of the average of number (resp. length) of limit cycles is in blue (resp. red).

2. Networks on 5 nodes

We performed 4000 experiments using 5-node networks. We varied the PF-distance from $d=0$ to $D=25\%$, 50% , 75% , 100% of the maximum distance of the network (Figure 4 and Table 2). We show the histograms for the slopes for the average number and length of limit cycles, and the median number and length of limit cycles.

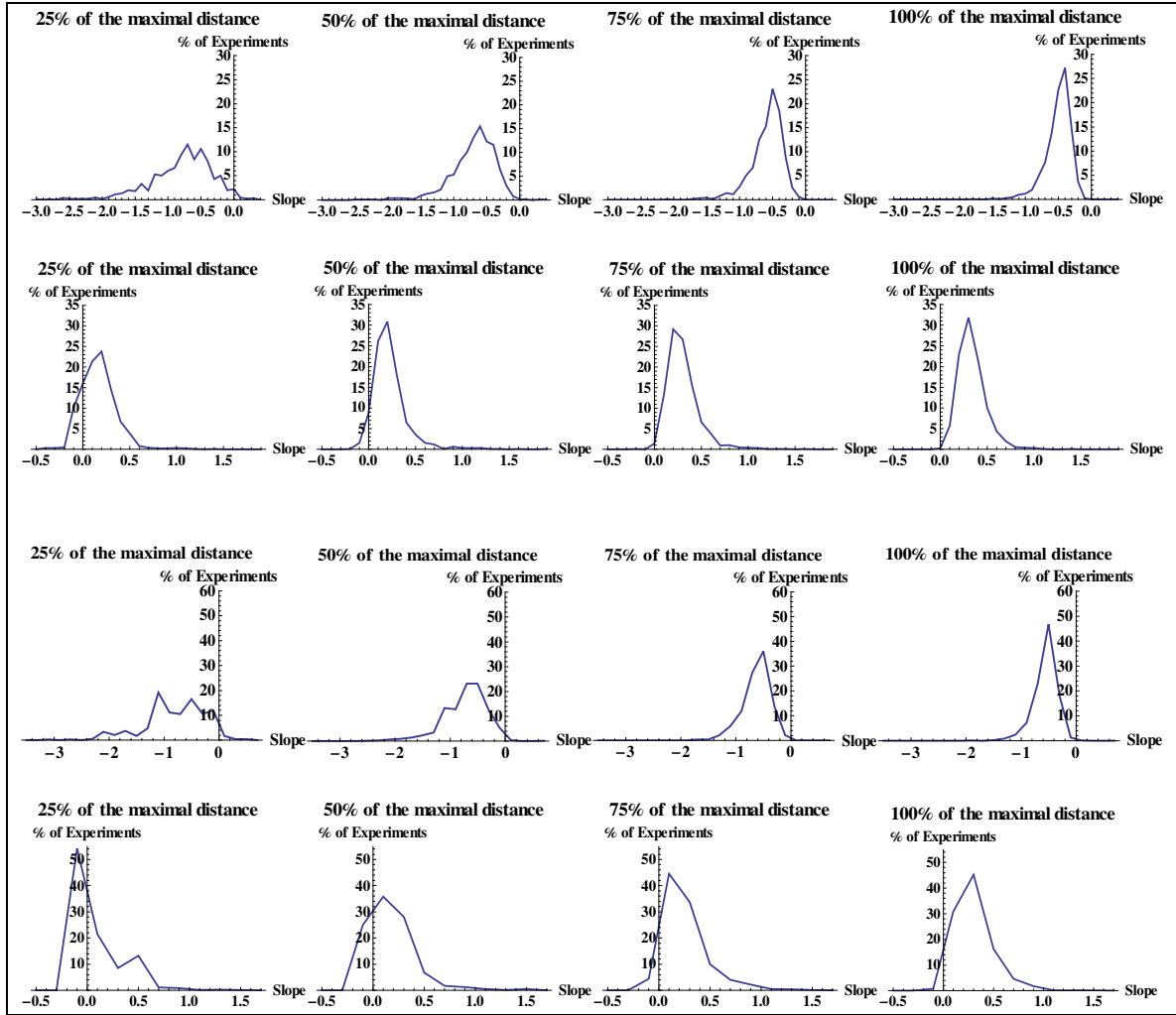


Figure 4. (**5-node networks**) Histograms for the average number/length of cycles (first/second row) and the median number/length of cycles (third/fourth row). The distance d varies from 0 up to $D=25$, 50 , 75 , 100% of the maximum distance in the first, second, third, fourth column, respectively.

D	Num. of Exp.	Av. Num.	Av. Len.	Med. Num.	Med. Len.
25%	1000	26	114	29	542
50%	1000	4	16	18	59
75%	1000	0	1	6	3
100%	1000	1	0	2	0

Table 2

We also performed 40000 experiments using 5-node networks, where we varied the PF-distance from $d=0$ to $D = D^* + \text{the number of self loops}$, where $D^*=2, 3, 5$, Maximum distance, respectively (Figure 5 and Table 3). We show the histograms for the slopes for the average number and length of limit cycles, and the median number and length of limit cycles.

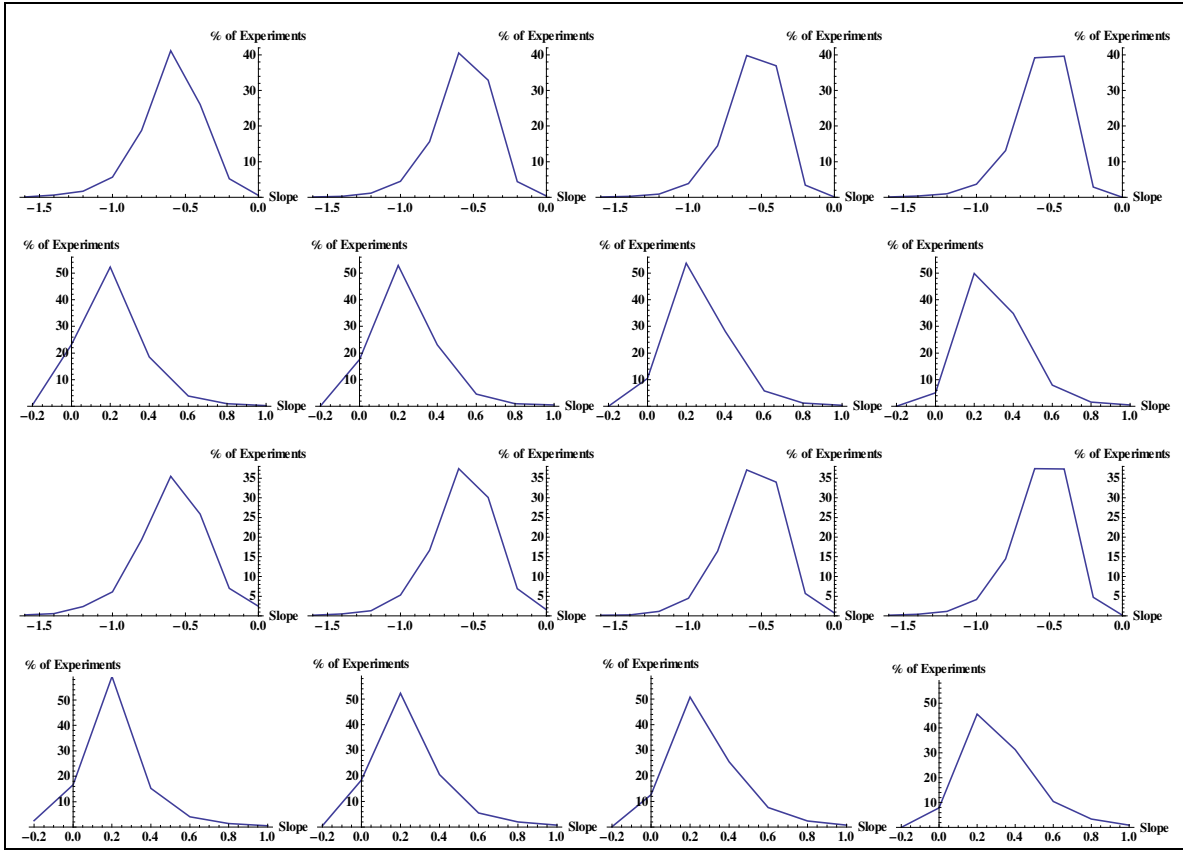


Figure 5. (**5-node networks**) Histograms for the average number/length of cycles (first/second row) and the median number/length of cycles (third/fourth row). The PF-distance varies from 0 to $D^* + \text{the number of self loops}$, where $D^*=2, 3, 5$, Maximum distance, respectively of the network without self loops.

D*	Num. of Exp.	Av. Num.	Av. Len.	Med. Num.	Med. Len.
2	10000	63	53	221	258
3	10000	33	19	153	44
5	10000	9	6	65	7
Max. Dis.	10000	6	5	54	7

Table 3

3. Networks of size $n=7, 10$

The computationally expensive part of the analysis made for 5-node networks is the computation of the maximal PF-distance of a given directed graph. Hence, we performed experiments using networks of $n=7$, $n=10$ where we considered the distance to vary from 0 to a fraction of the number of nodes. The results are shown in Table 4, 5, respectively. We show the histograms for the slopes for the average number and length of limit cycles, and the median number and length of limit cycles (Figures 6, 7).

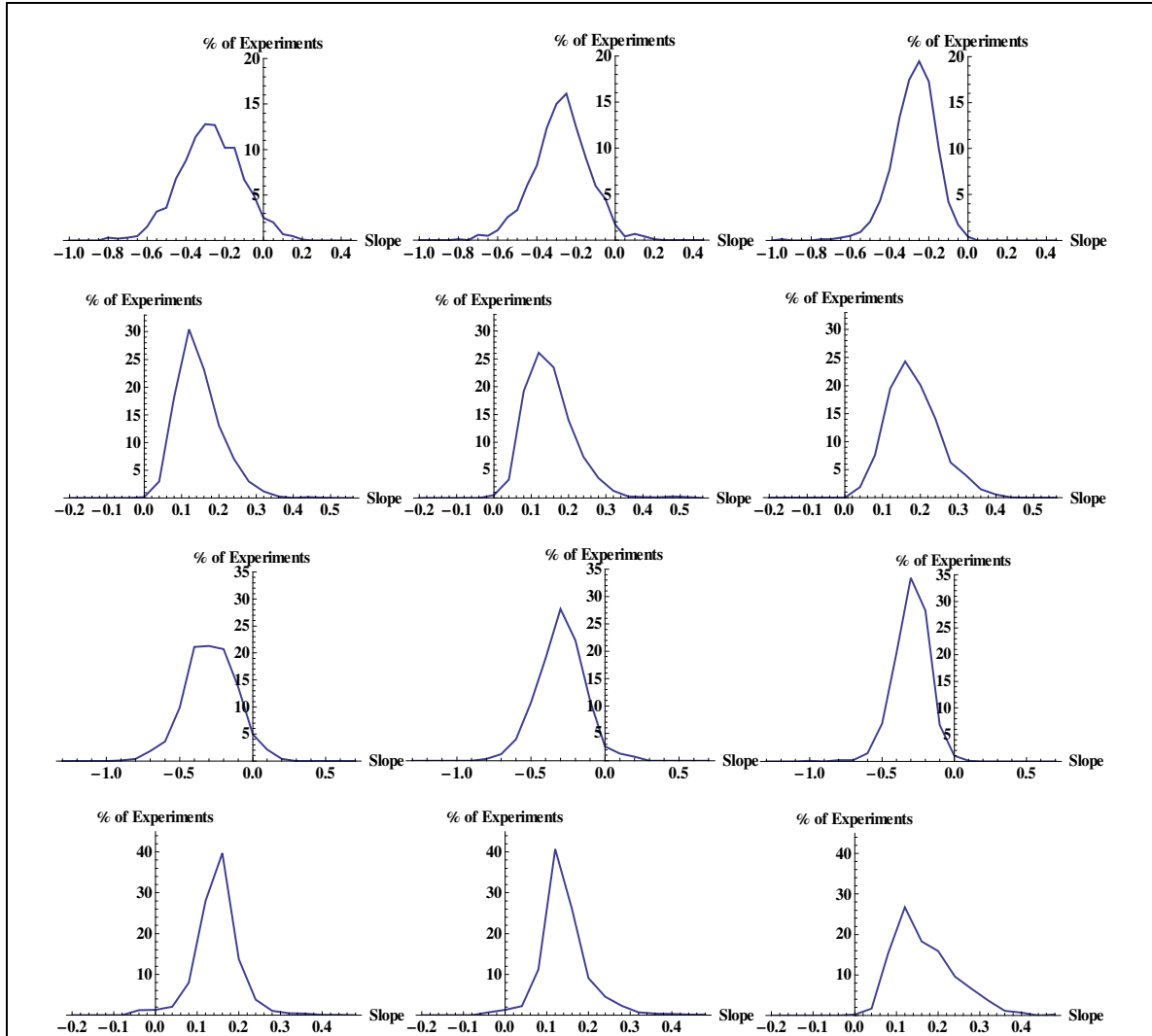


Figure 6. (**7-node networks**) Histograms for the average number/length of cycles (first/second row) and the median number/length of cycles (third/fourth row). The PF-distance varies from 0 to D^* +the number of self loops where $D^*=3,4,7$, respectively.

D^*	Num. of Exp.	Av. Num.	Av. Len.	Med. Num.	Med. Len.
3	1000	58	0	72	12
4	1000	33	0	44	8
7	1000	4	0	10	0

Table 4

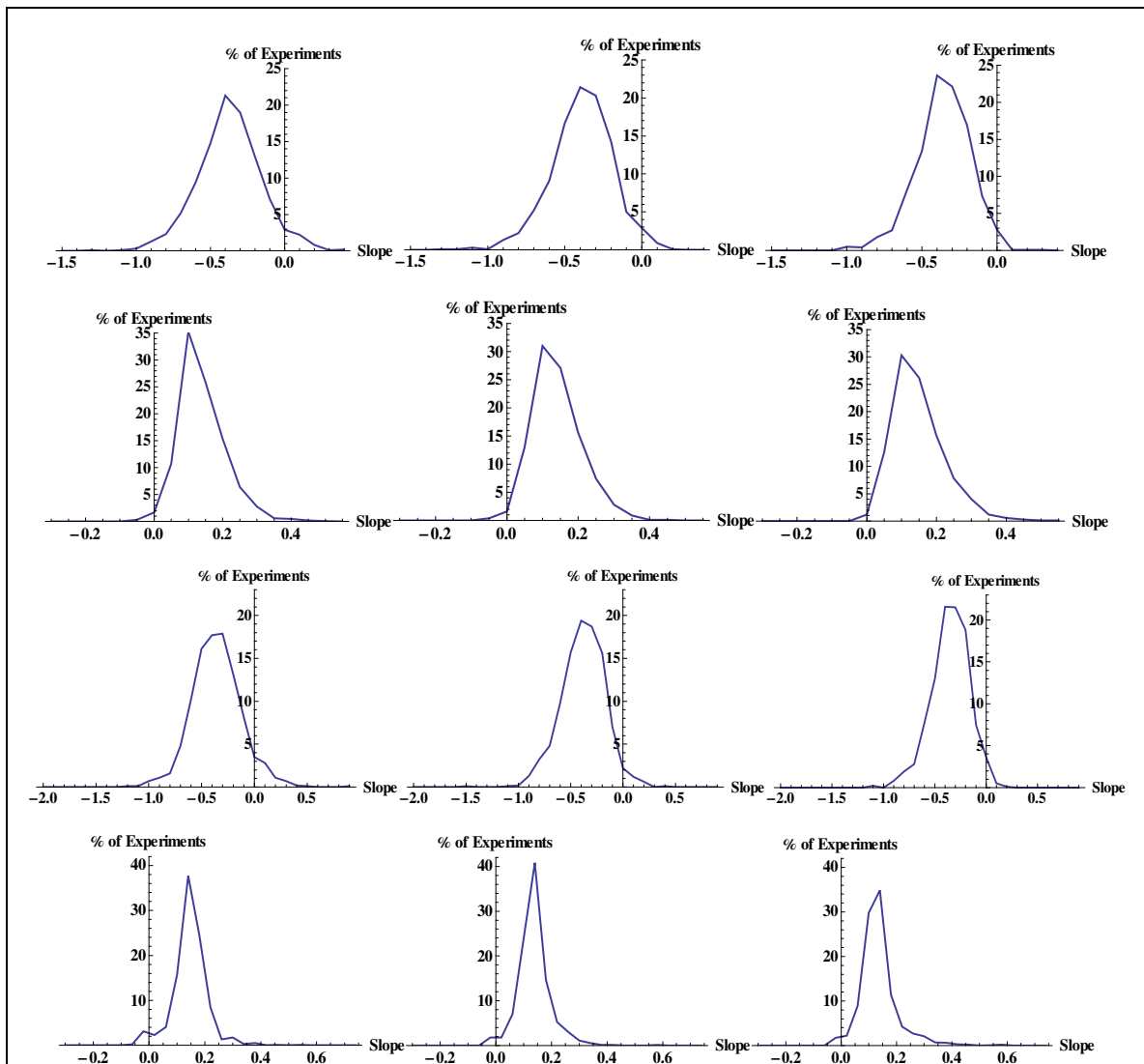


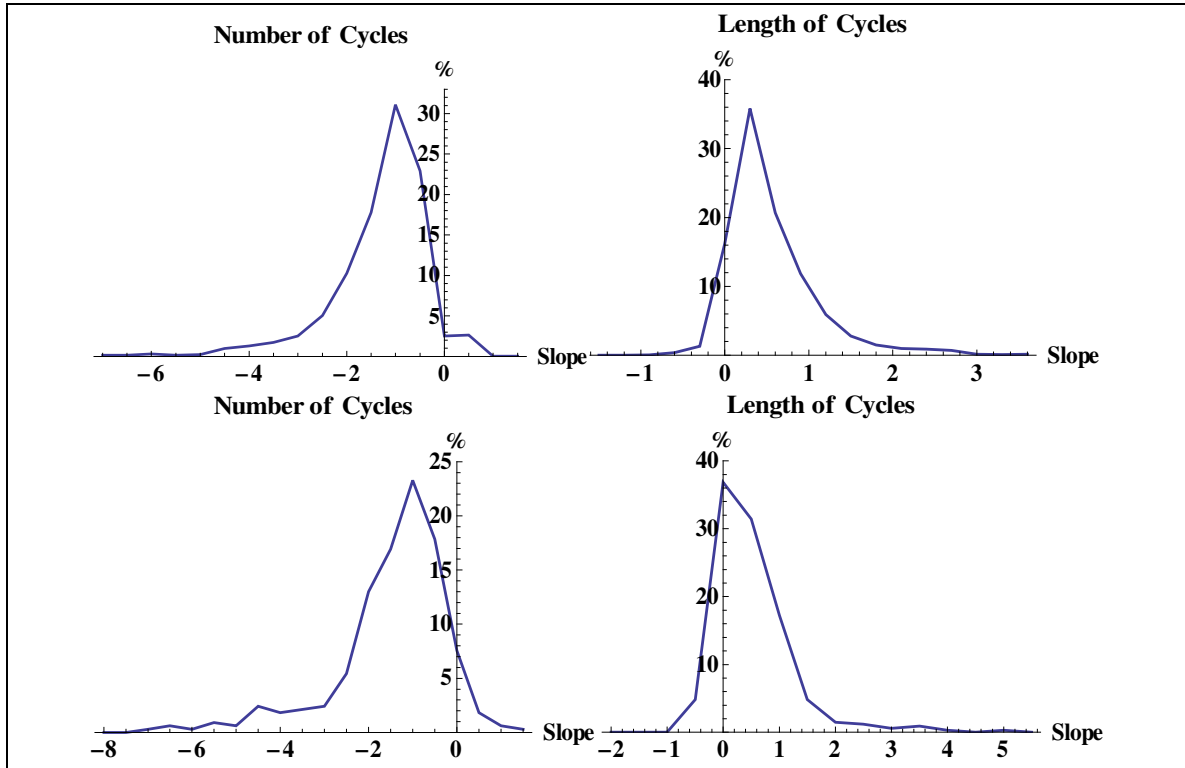
Figure 7. **(10-node networks)** Histograms for the average number/length of cycles (first/second row) and the median number/length of cycles (third/fourth row). The PF-distance varies from 0 to D^* +the number of negative self loops where $D^*=3,4,5$, respectively.

D^*	Num. of Exp.	Av. Num.	Av. Len.	Med. Num.	Med. Len.
3	1000	61	3	80	17
4	1000	40	4	40	13
5	1000	31	0	39	13

Table 5

4. Networks of $n=15$, $n=20$, $n=100$ nodes

We performed experiments with networks of $n=15$, 20, 100 nodes. The results are shown in Figure 8 and Table 6. The dynamics of Boolean networks on 100 nodes were computed using 10000 random initializations.



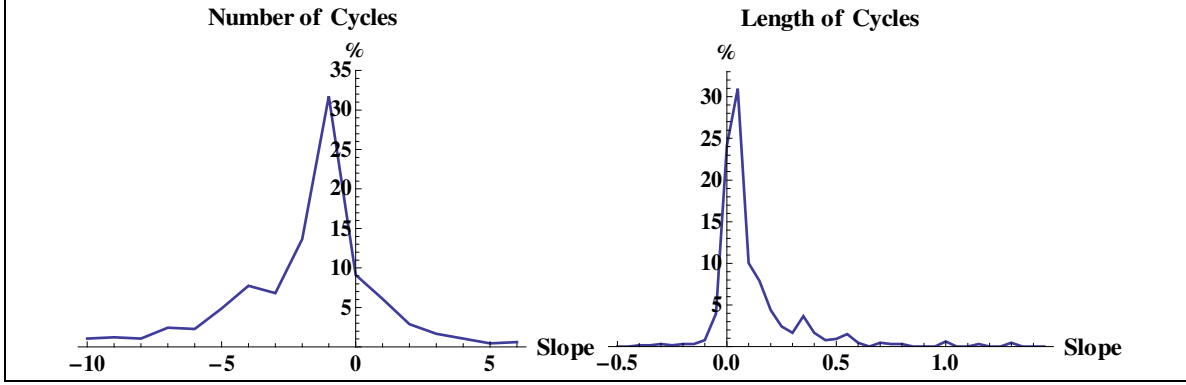


Figure 8. **(15, 20, 100-node networks)** Histograms for the average number and length of cycles for 15-node networks (first row), 20-node networks (second row) and 100-node networks (third row).

n	Num. of Exp.	Av. Num.	Av. Len.
15	2921	95.72	98.25
20	331	90.03	94.86
100	659	77.39	93.93

Table 6

5. An Algorithm for computing the maximal PF distance.

As we noted in body of the article, the distance of any graph is the sum of the distances of its strongly connected components. Furthermore, it is easy to see that each self loop with a negative sign increased the distance by 1. Therefore, without loss of generality, we assume the graph is strongly connected without self loops.

Input: The adjacency matrix $A=(a_{ij})$ of a strongly connected directed graph G on n nodes with m edges.

Output: The maximal distance D of G and a sign assignment S with exactly D negative signs and PF-distance D .

1. Let $B = \left\lfloor \sum_{i=1}^n \left\lfloor \sum_{j=1}^n (a_{ij} + a_{ji}) / 2 \right\rfloor / 2 \right\rfloor$,
2. For $D=B$ to 1, do
 - a. For $i=1$ to $\binom{m}{D}$ (there are $\binom{m}{D}$ sign assignments having d negative signs)
 - Let S be the i -th sign assignment of A
 - If the PF-distance of S equal to D , then RETURN (D,S)

This algorithm is guaranteed to return the maximal distance and a sign assignment of A with D negative signs and distance D because of the following Lemmas.

Lemma A. The PF-distance of S is less than or equal to

$$\left\lceil \sum_{i=1}^n \left\lfloor \sum_{j=1}^n (a_{ij} + a_{ji}) / 2 \right\rfloor / 2 \right\rceil$$

Proof. It follows from the fact that for any given vertex we can flip the sign of the incoming outgoing edges without changing the PF distance. Then we can obtain more positive than negative edges around every vertex (the dynamics of the Boolean network associated to the directed graph does not change either).

Lemma B. If PF-distance of S is d, then there exists a sequence of vertices, $v_1, v_2, \dots, v_t$ such that the sign assignment R obtained by changing the sign around those vertices has distance d with exactly d negative signs.

Proof. (By induction on d.) Recall that flipping the signs of the (in and out) edges of a given vertex in S does not change its PF-distance [24]. For d=0, the network is monotone and hence, by definition, the statement follows.

6. An Algorithm to check if a network is a PF network.

The following algorithm checks whether a network is positive feedback, which is equivalent to check if there are no negative feedback loops. Again, without loss of generality, we assume the graph is strongly connected without self loops.

Input: The adjacency matrix $A=(a_{ij})$ of a signed, strongly connected, directed graph G that has no self loop.

Output: TRUE if the G is a PF network and FALSE if G is not.

1. Let P and N be (0,1)-matrices such that $A=P-N$.
2. Let $P'=P$, $N'=N$.
3. For $i=1$ to n , DO

If N' has a nonzero diagonal entry,

RETURN FALSE

Else, let $P'=P'P+N'N$ and $N'=N'P+P'N$

4. RETURN TRUE

At each step i, P' keeps track of the number of positive paths and N' keeps track of the number of negative paths of length i. Then, the diagonal entries of N' correspond to the number of negative closed paths (any negative feedback loop will appear as a nonzero diagonal element of N').